Instagram Virality - Proposal

Problem Description

In the past decade social media has become one of the greatest drivers of culture. What goes viral on social media is highly impactful to society and what is popular. Instagram is one of the largest of the social media platforms, making it one of the most influential drivers of culture. In the past decade, vertical short-form video platforms, such as Instagram Reels, have become one of the most popular ways of consuming social media content. Being able to go viral on social media is by extension the ability to shape culture. This is why it is important to determine which characteristics of an Instagram Reel are most associated with virality, and how does observing these characteristics change the probability that a Reel is viral? This is important because it shapes generations' culture and shows what is valued in our society.

DataSet

We intend to use approximate inference on the What Makes an Instagram Reel Go Viral Dataset from Kaggle. This dataset includes 100,000 synthetic samples and 26 features including posting time, video length, virality score, and more.

Methodology

First we will figure out which features from the dataset we think will be relevant and which are not relevant to answering our question (ex. reel_id, creator_id). With said relevant features we will make an approximate Inference Bayesian Network using that subset of the 26 features.

Then we will turn the continuous variables into discrete ones (turn everything into binary variables). For example we will make the top 0.5% highest virality scores 1 and the rest 0.

We will start with a naive network based on reasonable assumptions like in figure (1). We will give reasonable estimates for the CPTs. Using this Bayesian Network, we will perform inference to answer questions such as how the probability of virality changes given evidence about features such as an engaging hook, number of impressions, number of hashtags, or account popularity. We will use an approximate inference method such as likelihood weighting and compare the estimated probabilities using different numbers of samples.

We then will experiment with an alternative Bayesian network such as figure (2) to explore how other assumptions of independence impact our inference calculations.

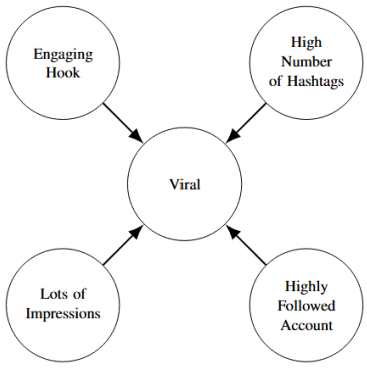


Figure 1: Initial naive Bayesian Network where all selected features directly influence virality.

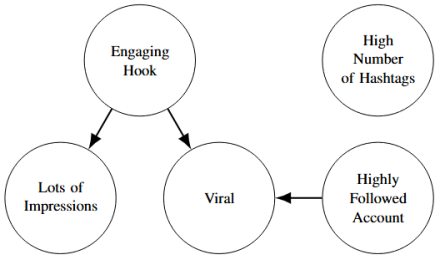


Figure 2: Alternative Bayesian Network with conditional relationships between Reel features.